

ELISAVETA (LISA) SAMOYLOV

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA

Expected December 2027

Masters in ML

Dartmouth College, Hanover, NH

Graduating June 2026

Bachelor of Arts: Majors in Mathematics and Computer Science

Honors/Awards: Rufus Choate Scholar, Lovelace Scholar, Neukom Scholar, Leslie Center for Humanities Fellowship

Citations (Top 2% of the class) : Machine Learning, Artificial Intelligence, Advanced Natural Language Processing

PUBLICATIONS (STARRED PAPERS LISTED ALPHABETICALLY. OTHERS LISTED IN ORDER OF CONTRIBUTION AS PER CUSTOM IN MATHEMATICS)

Modeling Tactics as Operators: Effect-Grounded Representations for Lean Theorem Proving

Elisaveta (Lisa) Samoylov, Soroush Vosoughi

[EMNLP \[MathNLP Workshop Oral and Poster\]](#)

*Properties and Comparisons of Various Graphs and Their Codes

Andreas Garcia, Layla Jarrahy, **Elisaveta (Lisa) Samoylov**

[IEEE MIT Undergraduate Research Technology Conference \(URTC\)](#) [Talk]

Proofs as Trajectories: Learning Lean Proof-Step Representations from State Changes

Elisaveta (Lisa) Samoylov, Soroush Vosoughi

Under review. (ACL Student Research Workshop)

Manifold Word Embeddings

Ankita Sarkar, Elisaveta (Lisa) Samoylov, Soroush Vosoughi

Under review. (TMLR)

*New Quantum Codes from Generalized Toeplitz Graphs

Andreas Garcia, Layla Jarrahy, **Elisaveta (Lisa) Samoylov**, Padmapani Seneviratne

Under review. (Discrete Mathematics)

WORK EXPERIENCE:

Machine Learning Engineer Intern -- KUNGFU.AI, Austin, TX

Jan – May 2026

Actor-Critic Reinforcement Learning: Analyzed hallucination patterns in schema-guided LLM outputs and implemented RL-based mitigation strategies to reduce schema violations relative to an ACL '25 baseline. Validated improvements through multi-GPU distributed training and schema-specific evaluation metrics

TEACHING EXPERIENCE

COSC30: Discrete Mathematics

COSC31: Algorithms

COSC74: Machine Learning

COSC76: Artificial Intelligence

COSC89: Advanced Natural Language Processing

Math31: Topics in Algebra

POSTERS AND TALKS

MathNLP Workshop Oral and Poster at EMNLP

November 2025

Modeling Tactics as Operators: Effect-Grounded Representations for Lean Theorem Proving

JMM AMS - PME Undergraduate Student Poster Session, I

January 2025

Quantum error-correcting codes from generalized Toeplitz graphs

MIT URTC Oral Presentation Track #1A

September 2024

Properties and Comparisons of Various Graphs and Their Codes [Top 5 Paper]